QUEZON CITY UNIVERSITY

SAN BARTOLOME | SAN FRANCISCO | BATASAN HILLS

COLLEGE OF COMPUTER STUDIES
Computer Science Department

COURSE SYLLABUS

COURSE TITLE	Advanced Database Systems
COURSE CODE	IM101
CREDIT UNITS	3 units Lecture & Laboratory
COURSE PREREQUISITE	CC105
COURSE DESCRIPTION	This course focuses in Advanced SQL using Oracle application including the Oracle database architecture, installation, managing database storage structure and user security. It also provides a foundation for database administration, data warehousing and data mining. This will help the students to develop their database skills that are essential to good programming practices and problem-solving skill.

VISION	MISSION	CORE VALUES
To be recognized as the #1 local university of employable graduates.	*To provide a comprehensive education that enhances the lives of QCU students for nation building and as world citizens.*	Jointness of undertaking and Organizational adaptability with Yoke of Efficiency and Effectiveness
INSTITUTIONAL OUTCOMES	**PROGRAM OUTCOMES**	**COURSE OUTCOMES**
QCU graduates are: Qualified in their chosen field locally and globally Competent and effective communicators Unique in demonstrating their sense of genuine nationalism and patriotism	Common to all Programs The graduates of have the ability to: a.)Articulate and discuss the latest developments in the specific field of practice; b.)Effectively communicate orally and in writing using both English and Filipino c.)Work effectively and independently in multi-disciplinary and multi-cultural teams	By the end of the course, students should be able to: LO1. Analyze an existing database system with respect to quality issues. Reliability, scalability, efficiency, effectiveness and security LO2. Design a database based on user requirements using a widely used modeling notation, and able to use declarative query language to elicit information.

	d.)Act in recognition of professional, social, and ethical responsibility e.)Preserve and promote "Filipino historical and cultural heritage" Common to the discipline The graduates of BSCS must have the ability to: a.)Analyze complex problems, and identify and define the computing requirements needed to design an appropriate solution b.)Apply computing and other knowledge domains to address real word problems c.)Design and develop computing solutions using a system-level perspective d.) Utilize modern computing tools	

OUTCOMES-BASED TEACHING AND LEARNING (OBTL) PLAN

WEEK	LEARNING OUTCOMES	TOPIC	METHODOLOGY	LEARNING AND TEACHING RESOURCES	ASSESSMENT and TASKS
1	• Explain the mission and vision of QCU. • Analyze how the study of Advanced Database Systems shares achieving the vision and mission of the university.	**Course Orientation** • Mission • Vision • Course Syllabus • Grading System • Classroom policies • Course Requirements	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	Oral Recitation/Selected Responses
2	• Identify the terminologies and	Review of Database System Concepts	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-	Oral Recitation Assignment – Research about

WEEK	LEARNING OUTCOMES	TOPIC	METHODOLOGY	LEARNING AND TEACHING RESOURCES	ASSESSMENT and TASKS
	basic concepts of Database Systems. • Understand the Database Environment. • Enumerate and discuss the characteristics of business rule. • Define the concepts of data models, schema and instances.	• Basic Terminologies • Database System Environment • Business Rule • Data Models, Schema and Instances		XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	Enhance Entity Relationship Diagram
3	• Define sub class, superclass and inheritance. • Discuss the various types of constrains on specialization/generalization. • Identify the different categories in	Enhanced Entity Relationship Diagram • Sub classes, superclass and inheritance • Specialization and Generalization • Constraints and characteristics of specialization and generalization hierarchies	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	Oral Recitation/Brainstorming Laboratory Activity – Creating ERD Diagram of a simple transaction

WEEK	LEARNING OUTCOMES	TOPIC	METHODOLOGY	LEARNING AND TEACHING RESOURCES	ASSESSMENT and TASKS
	modeling the UNION.				
4	• Discuss the major architectural components of Oracle Database. • Identify the task of an Oracle database administrator. • Understand how to install Oracle Database.	Introduction to Oracle Database • Exploring the Oracle Database Architecture - Oracle Database - Connecting to a server - Oracle Database Server Architecture • Installing your Oracle Software - Task of an Oracle Database Administrator - Planning your installation - System requirement	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	Can identify the different architectural components of Oracle Database. Can properly install Oracle Database. Project Proposal Quiz # 1 (Lesson 2-4)
5	• Discuss how table data is stored in oracle database. • Identify the user security in oracle database. • Understand the importance of user accounts, roles and privileges.	Managing Database Storage Structure and User Security • Managing Database Storage - How Table is stored - Table space • User Security - Database User Accounts - Privileges - Roles	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	Can create database connection and modify data table using SQL Developer. Can Run SQL Statements in SQL Developer Laboratory Activity –
6	• Discuss how to	Oracle SQL Developer and	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/dri	Can create database connection

WEEK	LEARNING OUTCOMES	TOPIC	METHODOLOGY	LEARNING AND TEACHING RESOURCES	ASSESSMENT and TASKS
	create and modify a database connection using SQL Developer. • Demonstrate how to run SQL statements in SQL Developer.	other tools • Working with SQL Developer - Start and stop database service • SQL Developer to work with a database - Create a database connection - Export or import database connection - Navigate through database objects - View the column definitions for a table - View the data for a table - Edit the column definition • SQL Developer to Run SQL statements - Enter and execute a SQL statements - Handle syntax errors - Open and save SQL		ve/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	and modify data table using SQL Developer. Can Run SQL Statements in SQL Developer. Laboratory Activity - Create a database and Student Table using Oracle Database

WEEK	LEARNING OUTCOMES	TOPIC	METHODOLOGY	LEARNING AND TEACHING RESOURCES	ASSESSMENT and TASKS
		Statements			
7	• Discuss how to retrieve data in SQL Developer using select statement and where clause. • Demonstrate how to retrieve data in SQL Developer using select and where clause.	**Query Processing** • Select Statement • Where Clause	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	Can retrieve data in SQL Developer using select and where clause. Quiz # 2 (Lesson 5 - 7)
8	• Identify the different types of Joins. • Demonstrate how to retrieve data in SQL Developer using different types of joins.	Query Processing cont. • Inner join • Outer join • Union	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	Can retrieve data in SQL Developer using different types of join. Present 50% of Project Laboratory Activity
9	MIDTERM EXAMINATION				
10	• Discuss how to run Subqueries in SQL Developer. • Understand where and when to use subqueries. • Demonstrate subqueries in SQL	Sub queries • Introduction to subqueries - How to use subqueries - Subqueries compare to joins • Subqueries in search condition - Subqueries in IN operator - How to use ALL keyword	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing	Can run subqueries in SQL Developer Laboratory Activity

WEEK	LEARNING OUTCOMES	TOPIC	METHODOLOGY	LEARNING AND TEACHING RESOURCES	ASSESSMENT and TASKS
	Developer.	- How to use ANY and SOME - How to use EXIST operator		E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	
11	• Understand the roles of data and database administration. • Discuss the database environment's human architecture. • Identify the different types of data and database security.	Database Administration • The Roles of Data and Database Administration • The Database Environment's Human Component	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	Can differentiate data and database administration. Assignment
12	• Discuss the database environment's human architecture. • Identify the different types of data and database security	Database Administration cont. • Security - Data Security - Database Security - Security Policies - Security Vulnerabilities	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	Can give real life situation in relation to database security. Quiz Assignment
13	• Discuss the basic	Transaction Management	Blended Learning	Powerpoint Slide Powerpoint	Can give examples of transaction

WEEK	LEARNING OUTCOMES	TOPIC	METHODOLOGY	LEARNING AND TEACHING RESOURCES	ASSESSMENT and TASKS
	concept of transaction processing. • Define the properties of transactions.	• Introduction to Transaction Processing • Transaction and System concepts • Desirable properties of transactions		Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QgCZEQ0rTzkC?usp=sharing	processing system. Laboratory
14	• Understand the basic concepts and application of data warehousing and data mining.	Introduction to Data warehousing and Data mining • Overview of Data warehouse - Introduction, definitions and terminology - Characteristics of data warehouses - Data modeling for data warehouses - Building a data warehouse - Typical functionality of a database warehouses.	Blended Learning	Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3olhUvz_TuC7QgCZEQ0rTzkC?usp=sharing	Can create a data model of data warehouse. Laboratory Activity
15		Introduction to Data warehousing and Data mining • Data mining concepts - Overview of data mining technology		Powerpoint Slide Powerpoint Link:https://drive.google.com/drive/folders/1K-XmtSA722_yNorr8EeomW1_jbWzvYLt?usp=sharing	Can identify the different approaches in data mining. Quiz

WEEK	LEARNING OUTCOMES	TOPIC	METHODOLOGY	LEARNING AND TEACHING RESOURCES	ASSESSMENT and TASKS
		- Association rules - Classification - Clustering - Approaches to other data mining problems - Application of data mining		E-Book Link:https://drive.google.com/drive/folders/1jPek5K0v3oIhUvz_TuC7QqCZEQ0rTzkC?usp=sharing	
16	• Present and evaluate the final course project.	**Project Development**	Blended Learning	Powerpoint Slide Powerpoint Projector Program	Demonstration Present 70-90% of Project
17	• Present and evaluate the final course project.	**Project Presentation**	Blended Learning	Powerpoint Slide Powerpoint Projector Program	Present 100% of project
18	FINAL EXAMINATION				

SUGGESTED READINGS AND REFERENCES

In APA 7th Ed arrangement/listings

REFERENCES:

Marek Chmel, Vladimír Mužný (2020), SQL Server 2019 Administrators Guide A definitive guide for DBAs to implement, monitor, and maintain enterprise database, Packt Publishing
Abraham Silberschatz, Henry F. Korth etc.(2020), 2020_Database System Concepts, McGraw-Hill Education
Andreas Meier, Michael Kaufmann (2019), SQL NoSQL Databases Models, Languages, Consistency Options and Architectures for Big Data Management, Springer Vieweg, Cengage
Carlos Coronel Steven Morris(2019), Database Systems Design, Implementation, and Management, Cengage
GoalKicker.com (2018), Oracle Database Notes for Professionals, Oracle® Database group(s), company(s)

COURSE POLICIES AND STANDARDS

Expectation from Student

The student's responsibility is to come to each class prepared, and to participate actively in the discussions. Student should read the assigned coursework prior to class, and to present reports as scheduled. S/he is also expected to take all quizzes and examinations and to submit all course requirements on the dates scheduled. S/he is expected to observe proper and polite behavior at all times, adhering to the rules and policies of QCU.

Language of Instruction
The language of instruction is English.

Attendance
Students are required to attend all classes starting with the first meeting. Non-attendance in any required class or academic activity constitutes an absence. A student who has been absent for more than 20 percent of the hours of recitation, lectures, or any other scheduled work in this course shall be automatically dropped from the class roll and the Registrar shall be advised accordingly.

Student Rights and Responsibilities
The QCU Student Manual establishes students' freedoms and protections as well as expectations of appropriate behavior and ethical academic work. The Handbook includes items such as the Policy on Student Rights, the Policy on Student Conduct, and the Academic Integrity Policy.

Academic Integrity
Any work that students submit should be their own work (i.e., not borrowed/copied from any other source, including their classmates). When using other people's ideas to substantiate their ideas, students are expected to properly cite the original source. Proper citation procedures shall be discussed in class. Any act of cheating or plagiarism shall be treated in accordance with the Institute's Policy on Academic Integrity. Depending upon the individual violation, students could face penalties ranging from failing the assignment to failing the class.

Grading System
The College of Computer Science and Information Technology follows the university grading system given below:

Middle term Grade (MTG)		**Final Term Grade (FTG)**		**FINAL GRADE:**		
*Class Standing	60%	*Class Standing	60%	Midterm Grade	50%	
Midterm Exam Grade	40%	Final Exam Grade	40%	Final Term Grade	50%	
Total	***100%***	***Total***	***100%***	***Total***	***100%***	***or (MTG + FTG)/2***

**Class Standing (60%)*

- *Online Discussion/ Presentation 5%*
- *Assignment 10%*
- *Quizzes 25%*
- *Output/Machine Problem 20%*

Rubrics for Course Output/Requirements

Each Project group will be graded by the following:

Peer Evaluation	-	*20%*
Individual Grade	-	*30%*
Group Grade	-	*50%*
	=	*100%*

Student Course Portfolio
All exercises, activities and requirements for the course must be compiled by the students and must be submitted to the instructor before the end of the semester.

Learners with Disabilities
The University and the College of Computer Science and Information Technology strives to make all learning experiences as accessible as possible. If a student anticipates or experiences barriers based on their disability (including mental health, chronic or temporary medical conditions), the students need to inform their instructor immediately so that they can discuss options privately. To establish reasonable accommodations, the students need to register in the "List of Learners with Disabilities" at the Office of Student Affairs and Services.

Syllabus Flexibility
The instructor reserves the right to amend or change this syllabus as needed. For example, the instructor may modify content, and/or substitute assignments in response to institutional, weather, or class situations. Any modification, however, must be communicated by the instructor to the Dean through the Chairperson. Approved changes shall be discussed by the instructor in class.

EFFECTIVITY AND REVISION		PREPARED BY	REVIEWED BY	Checked by	RECOMMENDING APPROVAL	APPROVED BY
Issue No.	*Revision No.02*	Manandeg, Mary Anne B.				
QA Doc No.	*Effectivity* AY 2024-2025	Jayobo, Mary Jean Santos, Richard M. Name of Faculty	Christian B. Escoto, MSCS Program Chair Computer Science Department	Ms. Edlyn O. Aquino Director for Library Resources and Services	Isagani M. Tano, PhD-ELM, DIT Dean, College of Computer Studies	Dr. Bradford Antonio Martinez Vice President for Academic Affairs

The original copy of this document is filed at the Office of the Dean copy furnished the Quality Assurance Office.
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